Operational Context & Baseline Assumptions

SYNCOS ORCHESTRATION ENGINE

Explicit Assumptions

Operational & Fleet Context

- The facility operates continuously 24/7 across three standard production shifts.
- The mixed fleet consists exactly of 10 robots: 5 autonomous forklifts handling pallets and 5 dual-conveyor robots handling intermediate materials.
- SyncOS already possesses basic payload compatibility awareness, ensuring tasks are only assigned to the correct vehicle class.

The Orchestration Baseline

- SyncOS currently utilizes a strict First-In, First-Out (FIFO) logic, assigning tasks purely in the chronological order they arrive without weighting geographic proximity.

Platform & Technical Capabilities

- **Real-Time Spatial Awareness:** SyncOS maintains a real-time coordinate map of the facility, allowing it to mathematically calculate distances and track live X/Y coordinates for all vehicles.
- **Data Completeness at Source:** Customer systems (WMS, ERP, PLCs) transmit complete task parameters (pickup location, destination, payload type) at the exact moment a task is generated.
- **Fleet Command Authority:** SyncOS has the API authority to interrupt a robot mid-route and dynamically reassign it to a higher-priority task or force an idle robot to navigate to a charging station.
- **Existing Configuration Abstraction:** SyncOS possesses a core configuration layer, ensuring that new features can be toggled on or off per site without requiring a unique custom code build.

Environmental Factors

- **Shared Workspace:** The automated fleet operates in a shared physical environment alongside human workers and manual equipment, which can cause unpredictable physical interruptions and localized delays.